



describe a person:-

feature	Value
Weight	65
Height	170
Age	25
Salary	50000

Geo + Algebra $\Rightarrow [65, 170, 25, 50000]$

Netflix

Users	Action Movies	Romance	Horror
Arnold	9	1	2
Maharad	8	2	1

Recommend $\rightarrow$

A: H1 $\rightarrow$ 170, W1 $\rightarrow$ 70 $\rightarrow$ [170, 70]
 B: H2 $\rightarrow$ 171, W2 $\rightarrow$ 70 $\rightarrow$ [171, 70]
 close $\rightarrow$ similar

Vector dimension $\rightarrow$ number of features
 Num of dim $\rightarrow$ structural dimension
 [10, 20, 30] $\rightarrow$ Vector $\rightarrow$ 3
 nd array $\rightarrow$ 1

Scalar quantity $\rightarrow$ 5
 [1, 2]
 $5 \times [1, 2] \Rightarrow [5, 10]$

Vector magnitude

$$|\vec{v}| = \sqrt{x^2 + y^2 + \dots + z^2}$$

[3, 4]
 $|\vec{v}| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$

Salary $\rightarrow$ 50000 $\rightarrow$ [50000, 22]
 Age $\rightarrow$ 25 $\rightarrow$ [25, 22]
 Normalization
 Vectors to a common scale
 Unit vector normalization
 $\vec{v} \rightarrow$ unit vector ($\hat{v}$)
 $\hat{v} = \frac{\vec{v}}{|\vec{v}|}$
 $\hat{v} = \frac{[3, 4]}{5} = [\frac{3}{5}, \frac{4}{5}] = [0.6, 0.8]$

Dot Product

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3 + \dots + a_nb_n$$

[1, 2, 3] $\cdot$ [4, 5, 6]
 $a_1, a_2, a_3 \cdot b_1, b_2, b_3$
 $= 1 \times 4 + 2 \times 5 + 3 \times 6$
 $= 4 + 10 + 18$
 $= 32$

email converted to vector
 [buy, offer, money, meeting]
 spam email $\rightarrow$ [10, 8, 7, 0]
 normal email $\rightarrow$ [0, 0, 0, 9]
 classify $\rightarrow$ spam / not spam

S1 $\rightarrow$ Connect words with $\rightarrow$
 S2 $\rightarrow$ [10, 8, 7, 0] $\rightarrow$ spam
 S3 $\rightarrow$ [0, 0, 0, 9] $\rightarrow$ not spam
 S4 $\rightarrow$ New email $\rightarrow$ [5, 4, 3, 0]
 [10, 8, 7, 0] $\rightarrow$ 103 $\rightarrow$ very large
 [0, 0, 0, 9] $\rightarrow$ 0 $\rightarrow$ very small
 similarity $\rightarrow$ SPAM

Neural network
 $z = \omega \cdot x + b$
 ω weights vector
 x input vector
 b bias

Cosine Similarity

[1, 1] [100, 100]
 Distance huge
 focuses on direction
 $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$
 $\vec{a} \cdot \vec{b}$ dot product
 $|\vec{a}|$ length of vector a
 $|\vec{b}|$ length of vector b
 θ angle b/w vectors

$\vec{a} = (-1, +1)$
 $\vec{b} = (0.9, 0.5)$
 new email $\rightarrow$ 0.9 $\rightarrow$ same dir.
 spam email $\rightarrow$ 0 $\rightarrow$ unrelated
 $\rightarrow$ -1 $\rightarrow$ opposite

$\sin \theta = \frac{1}{\cos \theta}$
 $\cos \theta = \frac{1}{\sin \theta}$
 $\tan \theta = \frac{1}{\cos \theta}$
 $\sec \theta = \frac{1}{\cos \theta}$
 $\csc \theta = \frac{1}{\sin \theta}$
 sin cos tan
 P B P
 1 HB
 cos secant

Vector projection

$\rightarrow$ shadow of one vector onto another

A $\rightarrow$ direction vector
 B $\rightarrow$ actual vector
 projection $\rightarrow$ How much of B goes along A

projection of B onto A
 $\text{Proj}_A(B) = \frac{B \cdot A}{|A|^2} A$
 Dot product $\rightarrow B \cdot A \rightarrow$ similarity
 Divide by $|A|^2 \Rightarrow$ normalize direction
 multiply by A $\rightarrow$

Google $\rightarrow$ [10, 10, 10, 10] $\rightarrow$ 10 pages
 [10, 10, 10, 10] $\rightarrow$ 10 pages
 [10, 10, 10, 10] $\rightarrow$ 10 pages

Dot product
 $\vec{a} \cdot \vec{b} \rightarrow$ scalar quantity

Cross product

vector $\times$ vector
 $\vec{a} \times \vec{b} \rightarrow$ vector

$$\vec{a} \times \vec{b} = [a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1]$$

$\vec{a} = [1, 0, 0]$
 $\vec{b} = [0, 1, 0]$
 $\vec{a} \times \vec{b} = [0 - 0, 0 - 1 \times 0, 1 \times 1 - 0] = [0, 0, 1]$

$\vec{a} = [1, 0, 0]$ (x, y, z)
 $\vec{b} = [0, 1, 0]$ (x, y, z)
 $\vec{c} = [0, 0, 1]$ (x, y, z)